RESEARCH REPORT · VITKI

Fully tested finding

The runestones turned their text towards the road – half of the inscriptions within 45 degrees of the road, against a quarter expected

Viktor Kvant

Independent researcher · Vitki

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Timeline of the study: The analysis plan was locked 29 October 2026 · The analysis was run 29 October 2026 · The addendum plan for the sensitivity analyses was locked and run 31 October 2026 · A review before archiving found two coding errors 5 November 2026 · The check with corrected directions was run 5 November 2026

ABSTRACT

Background and aim. Where the runestones were raised has long been discussed, and the roads are one of the answers: stones by roads were seen. Single stones have been described as turned towards the road, but nobody has measured across the whole material which way the inscription was turned. The question here is whether the text was turned towards the road, towards the farm or towards a particular point of the compass.

Material. 219 records in the Swedish register of ancient monuments on 214 Swedish Viking-Age runestones that still stand in place and for which the register gives the direction the inscription faces. For 123 of the records there is a road to measure against, for 94 a farm site.

Method. A pre-registered study with four tests and Holm correction. The measure is the cosine of the angle between the inscription's direction and the direction to the road, compared with 9,999 shuffles of the directions within the province. Internal replication in two halves and ten sensitivity analyses under a locked addendum plan.

Results. Half of the stones, 50 percent, turn the inscription within 45 degrees of the road, against a quarter if the direction were random. Mean cos 0.37 against 0.01 expected, p = 0.0001, Holm-corrected p = 0.0004. It holds in both halves and in every sensitivity analysis, also without Uppland, parish by parish, with one stone per carver and without re-erected stones. Two coding errors found after publication do not change this; corrected, the association is slightly stronger (mean cos 0.39, 52 percent within 45 degrees; section 4.5). The directions do not differ detectably from an even spread, and we found no sign that the inscription was turned towards the farm (farm sites are known only from 17th–18th-century maps).

Conclusion. The text was turned towards those passing by on the road. The main weakness is that the

road in the register is often today's road; with only registered older routes and road markers the association is weak (28 percent within 45 degrees against 25 expected) but goes the same way.

Keywords: *runestones, Viking Age, roads, placement, orientation of the inscription, circular statistics, Swedish register of ancient monuments, pre-registration*

HIGHLIGHTS

- The runestones turned their text towards the road. We saw no consistent sign that the compass or the farm decided how they were turned.
- The carvers wrote for those who passed by: the stone was a message to travellers, not only a memorial by the home.
- It withstands every check we have, but the road in the register is often today's road, so the next test is the old maps.

1. Introduction

Gunnar Ekholm (1950) asked where the runestones were raised and answered that the stones of Uppland largely stand by the roads and waterways of the past, and that the stones at grave fields, too, are often explained by the road that passed the grave field. He noted that single stones are turned towards the road, at Bollsta in Vaksala (p. 138) and on Särsta land in Knivsta (p. 139), and that a re-erection can turn a stone: a stone on Ängby land near Knivsta has faced the railway since it was raised again in the 1860s (p. 140). Torun Zachrisson (1998) has related the stones to farm, boundary and grave field, and Lydia Klos (2009) has argued that most were raised at grave fields. Mats G. Larsson (2010, p. 252) has objected that the grave fields are thereby strongly overstated at the expense of other locations; he also notes that Klos used considerably more restrictive criteria than his own earlier survey for what counts as an ancient road (p. 250; see also McLeod 2014). Johan Bollaert (2016) compared stones along three Viking-Age roads with other stones in the same districts and found that the stones by the roads more often have striking ornament and more often belong to monuments of several stones, which supports that they were raised to be seen.

What is missing is a measure across the whole material of how the inscription itself was turned. Ekholm's observations concern single stones, and a stone may stand by a road but turn its text another way. The aim here is to test whether the inscription was turned towards the road, towards the farm or towards a particular point of the compass, with the information on the inscription's direction that the register gives for some stones.

2. Pre-registration: hypothesis and plan

The hypotheses, measures and rules of interpretation were written down in an analysis plan on 29 October 2026, before any direction had been computed. The plan has the SHA-256 checksum `cd73b962457fde0e…` (see Data and code). Before locking only the number of records was known: 456

records in the register give a direction, 369 concern a single stone with one direction, 219 of these stand in place, and 88 of the 369 mention a road with a compass point.

The plan tests four hypotheses with Holm correction: H1 that the directions are not evenly spread over the compass, H2 that the inscription is turned towards the road, H3 that it is turned towards the farm and H4 that bridge and road stones are turned towards the road more than others. A hypothesis *holds* when Holm p is below 0.05 and the direction is the same with p below 0.05 in both halves.

The plan did not contain the project's fixed sensitivity analyses, which are built for associations between two traits and not for directions. The corresponding steps for directions were therefore fixed in an addendum plan on 31 October 2026 (SHA-256 `4da9710de49fa2b6…`), before they were computed. Two of the steps, the halves and the source of the road, were already in the main plan. A check of re-erected stones was made after the main run on 29 October with a different word list; in the addendum plan the word list was locked before the step was computed.

3. Material

- **The stones.** Swedish Viking-Age runestones in the Scandinavian Runic-text Database (Rundata) that still stand in place and have a known location, and whose record in the Swedish National Heritage Board's register of ancient monuments concerns just that stone and gives one direction with the words "faces" and one of sixteen compass points: 219 records, which concern 214 stones (five stones have two records each and were counted twice in the locked analysis; see 4.5). Runestones only.
- **The road.** First the register's own statement of the stone's position relative to a road ("5 m E of road"; the road then lies in the opposite direction): 69 stones. Otherwise the nearest registered route or road marker within 500 metres. 123 stones with a road in all. The roads come from the National Heritage Board's register only.
- **The farm.** The nearest registered village or farm site within one kilometre: 94 stones.
- **The directions** are computed as bearings from the stone, with north as 0 degrees and the compass points in steps of 22.5 degrees.
- **The halves** are the project's fixed division parish by parish.

4. Method

4.1 The measure. For each stone the cosine of the angle between the inscription's direction and the direction to the road is computed: 1 when the inscription faces the road straight on, 0 when it faces sideways and −1 when it faces away. The mean over the stones is compared with 9,999 shuffles in which the inscriptions' directions are redistributed among the stones within the same province, so that each province's general pattern of directions is kept. The test is one-sided. The share of stones within 45 degrees of the road is also reported; if the direction were random it would be a quarter.

4.2 The other hypotheses. H1 is tested with Rayleigh's test of uniformity (Mardia & Jupp 2000). H3 is tested like H2 against the direction to the farm. H4 compares the mean cos towards the road between bridge and road stones and the others, with the label shuffled within the province, two-sided.

4.3 Correction and replication. Holm correction (Holm 1979) over the four tests. Each hypothesis is also tested in each half on its own.

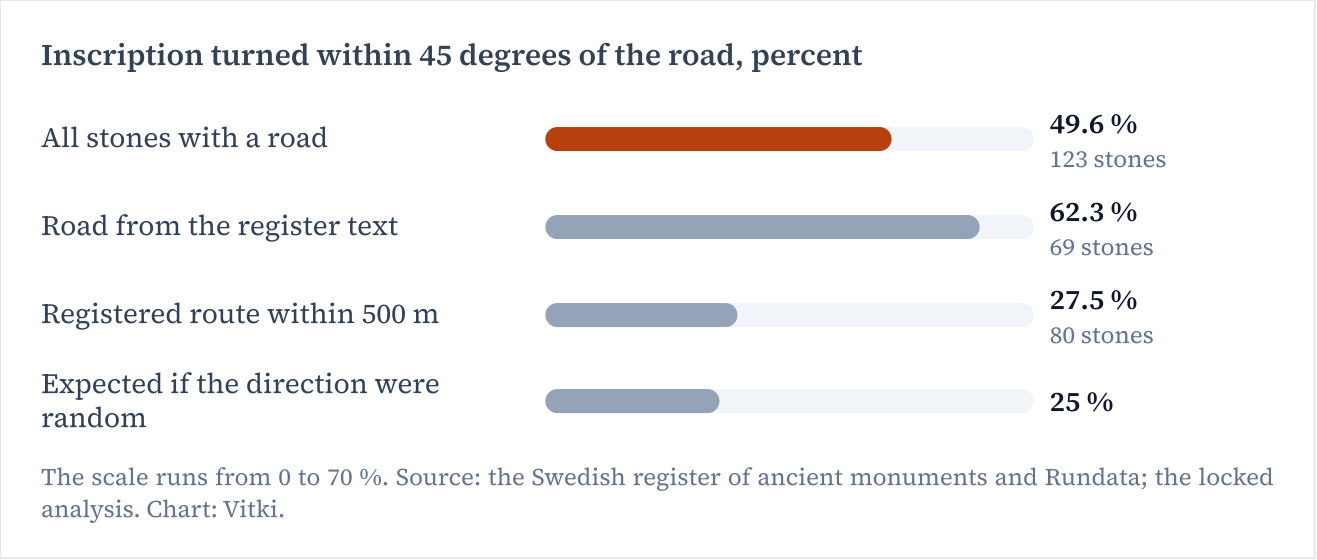
4.4 The sensitivity analyses under the addendum plan: the halves; without one province at a time (provinces with at least ten stones with a road; significance required only where at least two thirds of the stones remain); the parish as the unit, with one value per parish; one stone per secure carver; without re-erected stones; without stones the National Heritage Board's register Runor (2020) gives as moved; with the rule for moved stones after the correction of 27 October 2026 (see Data and code); only roads from the register text; and only registered routes and road markers. A step holds when the mean cos lies above the expected value and p is below 0.05.

4.5 Deviations from the plan. The analysis followed the plan, but a review before archiving (5 November 2026) found two coding errors in how the register text was read. The numbers in the results section are those of the locked analysis and have not been changed; the errors and a check made afterwards are reported here.

- **Gon qualifiers.** Nine road bearings and six inscription bearings with a gon qualifier ("V 40cg S", that is 40 gon or 36 degrees from west towards south; a gon is one four-hundredth of a full turn) were read as a single compass point. This concerns 9 of the 69 roads from the register text, with errors of 50–86 degrees, and 6 of the 219 inscription directions, with errors of 9–40 degrees. Corrected, the association is slightly stronger: mean cos 0.39 instead of 0.37 and 52 percent within 45 degrees instead of 50; with only roads from the register text 0.53 instead of 0.49.
- **Double records.** Five stones have two records each in the register and were counted twice; the 219 records concern 214 stones, and the 123 with a road concern 120. Counting each stone once, with corrected directions, gives mean cos 0.37 and 51 percent within 45 degrees.
- **The check.** With the corrected directions the main test and every sensitivity analysis were rerun with the same code and the same shuffles as under the plans; with the old directions the code reproduces the published numbers exactly. Every step holds, both with the 219 records and with each stone counted once. The main test gives p = 0.0001, half B p = 0.003 and 0.008 respectively, and the weakest step, only registered routes and road markers, p = 0.024 and 0.047. H1 and H3 do not change in substance (Rayleigh's test p = 0.46; towards the farm p = 0.29).

5. Results

5.1 Towards the road (H2). Of the 123 records with a road (120 stones; see 4.5), 50 percent turn the inscription within 45 degrees of the road. The mean cos is 0.37 against 0.01 expected, p = 0.0001, the lowest value 9,999 shuffles can give, and Holm-corrected p = 0.0004. Half A: 71 stones, mean cos 0.45, p = 0.0001. Half B: 52 stones, mean cos 0.26, p = 0.019. H2 *holds* by the plan's rule. In plain words: of the stones where we know where the road ran, half turn their text towards the road, within 45 degrees either way; if the direction were random, a quarter would.



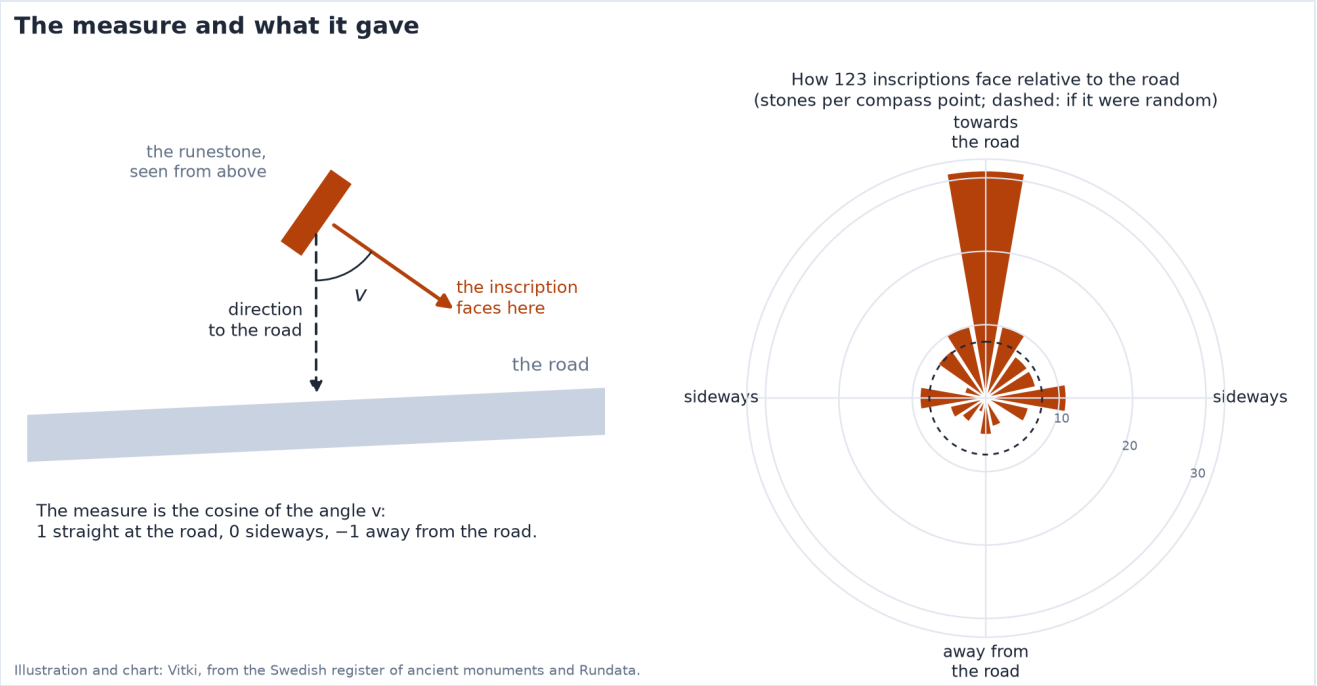


Figure 1. Left, the measure: the angle v between the direction the inscription faces and the direction to the road. Right, the 123 stones: 31 face the same compass point as the road (within 11.25 degrees), against about 8 if the direction were random. Illustration and chart: Vitki, from the Swedish register of ancient monuments and Rundata.

5.2 The compass, the farm and the bridges (H1, H3, H4). The directions do not differ detectably from an even spread over the compass: Rayleigh's test p = 0.49 (in half A p = 0.79, in half B p = 0.039: half B on its own leans towards the south-west, but neither half A nor the whole material does, so by the plan's rule it does not count). West is the most common (37 stones), then north (26), east (25) and south (25); 32 percent face somewhere between south-east and south-west, about what an even spread gives (31 percent, five of the sixteen points). Towards the farm: mean cos 0.01 against −0.02, p = 0.30; so we found no sign that the inscription was turned towards the farm, but the farm sites are known only from the maps of the seventeenth and eighteenth centuries. The bridge and road stones (9 with a road) do not differ detectably from other stones, p = 0.86, but with nine stones the test says little. None of the three hypotheses holds.

The plan's robustness tests. With the farm within 500 metres instead of one kilometre (52 stones): mean cos 0.01 against −0.06, p = 0.20. With only Uppland and Södermanland (92 stones with a road): mean cos 0.35 against −0.02, p = 0.0001, 52 percent within 45 degrees. The plan's two other tests, only roads from the register text and only registered routes and road markers, are among the sensitivity analyses in 5.3.

5.3 The sensitivity analyses. All hold:

- Half A (71 stones): mean cos 0.45 against 0.04, p = 0.0001.
- Half B (52 stones): mean cos 0.26 against 0.05, p = 0.016.
- Without one province at a time: without Uppland (62 stones left, half, so only the direction is required) mean cos 0.50, p = 0.0001; without Södermanland (92) 0.29, p = 0.0001; without Östergötland (113) 0.36, p = 0.0001.
- The parish as the unit, 76 parishes (123 stones): mean cos 0.36 against 0.02, p = 0.0001.
- One stone per secure carver, 20 carvers (93 stones): mean cos 0.38 against 0.03, p = 0.0001.
- Without re-erected stones, 14 removed (109 stones): mean cos 0.43 against 0.02, p = 0.0001.
- Without stones Runor gives as moved, 10 removed (113 stones): mean cos 0.35 against 0.01, p = 0.0001.
- With the rule for moved stones after the correction of 27 October 2026 none of the 123 counts as moved, since the main analysis is already built on that rule; the result is the same as the main test.
- Only roads from the register text, 62 percent within 45 degrees (69 stones): mean cos 0.49 against 0.07, p = 0.0001.
- Only registered routes and road markers, 28 percent within 45 degrees (80 stones): mean cos 0.16 against 0.01, p = 0.020.

The sensitivity analyses drew new shuffles, so the p values differ slightly from those of the main test: half B has p = 0.016 here and p = 0.019 in 5.1. The addendum plan's word list for re-erected stones hits 29 of all 219 stones; the check after the main run, with a wider word list, hit 58. Both give the same result.

6. Discussion

What the finding says. Those who raised the runestones turned the text towards the road. It holds in the whole material, in each half, without Uppland, parish by parish and with one stone per carver, so it does not depend on a few places or workshops. It also holds without the stones that have been re-erected, which could otherwise have been turned towards today's road, and when the coding errors in 4.5 are corrected. We found no consistent sign that the inscription was turned towards a particular point of the compass or towards the farm. That does not mean there was no such association: the farm sites are known only from the maps of the seventeenth and eighteenth centuries, and Rayleigh's test detects only a single preferred direction, not, for example, two opposite ones.

Against earlier research. The result gives a measure across the whole material to what Ekholm (1950) saw at single stones and to the idea that the stones by the roads were raised to be seen (Bollaert 2016). It does not contradict that many stones were raised at grave fields (Klos 2009): a stone can stand at a grave field and turn its text towards the road past it, as Ekholm (1950, p. 138) described the stone at Bollsta. The waterways are not measured here; McLeod (2014) shows that U 170 turns its text towards a former bay and has the road behind it.

Weaknesses, in order of weight.

1. **The road is often today's road.** The register text gives the road at the time of the survey. The association is strongest with roads from the register text (62 percent within 45 degrees) and weak with registered older routes and road markers (28 percent against 25 expected, p = 0.020). If roads were later laid past the stones, or stones were reset to face a newer road, the association with today's road may be stronger than with the Viking-Age one.
2. **The selection.** Only 214 stones in place have a single direction in the register, and we do not know what decided which stones received such a statement.
3. **Re-erections.** The word search does not catch every stone that has been re-erected, and a re-erection can turn a stone (Ekholm 1950, p. 140). The result is, however, stronger without the stones it finds, not weaker.
4. **The farm.** The farm sites are known only from the maps of the seventeenth and eighteenth centuries, and the Viking-Age farms may have lain elsewhere. That we found no association with the farm therefore says less than the association with the road.
5. **The resolution.** The directions are given in sixteen compass points, that is in steps of 22.5 degrees.

What would overturn the finding. If the association disappeared when the roads were measured on the oldest land-survey maps, or if a new survey of more stones' directions gave an even distribution relative to the road.

7. Conclusion

The runestones turned their text towards the road: half of the inscriptions face within 45 degrees of the road, against a quarter if the direction were random, and the association withstands every sensitivity analysis, also when the coding errors are corrected. The carvers wrote for those who passed by. The next step is to measure the roads on the maps of the seventeenth and eighteenth centuries.

Data and code

The analysis was made in Vitki 3.0.0 (git tag v3.0.0, 2026-10-20), an open research tool for runestones (vitkirunor.se), with the changes made after that version. Code state: the main analysis in commit 74ca1ef (2026-10-29), the sensitivity analyses in d0f5a6b (2026-10-31) and report version 2 in 4f80a63 (2026-11-01); the check in 4.5 was added with report version 3 in commit 6eae0af (2026-11-

05), together with the builder of that version. All code, the analysis plans and the result files are openly available on GitHub under GPL-3.0.

- The corpus: Rundata, frozen for the analysis as corpus `3f58e9339191761e` (2026-10-28), with the rule for moved stones after the correction of 27 October 2026 (commit 07d2ec0). The step with that rule in 5.3 reads corpus `e83202e01a5e6d27`, which has the same rows.
- The analysis plans: `analysplaner/2026-10-29-ristningens-riktning.md` and `analysplaner/2026-10-31-riktning-trappan.md`, with their checksums in the `.sha256` files beside them.
- The analysis: `utdata/riktning/analys.py` (results `riktning.json`, the stones in `stenar.csv`); the sensitivity analyses `utdata/riktning/trappa.py` (`trappan.json`).
- The check in 4.5: `utdata/riktning/gon_kontroll.py` (`gon_kontroll.json`) reads the gon qualifiers, counts each stone once and reruns every sensitivity analysis; with the old directions it reproduces `trappan.json` exactly.
- The register: the Swedish National Heritage Board's register of ancient monuments via K-samsök (CC0); the roads come from there only. The files in `data/raa/` are not under version control but are fetched with `scripts/fetch_kmr_runstenar.py` (the runestone records, fetched 2026-10-29), `scripts/fetch_raa_remains.py` (routes, 2026-10-12) and `scripts/fetch_raa_extra.sh` (road markers and farm sites, 2026-10-29); Runor with `scripts/fetch_runor.py` (2026-10-16). The checksums of the fetched files are in `gon_kontroll.json`.
- The report is built by `utdata/blogg/build/build_rapport_riktning.py`, which reads every number from the result files and stops if anything disagrees.

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University.

Change log

- **Version 1, 2026-10-31.** First published version.
- **Version 2, 2026-11-01.** Figure 1 added: a sketch of the measure and a rose diagram of how the inscriptions face relative to the road. No numbers, interpretations or conclusions have changed.
- **Version 3, 2026-11-05.** Corrections after a review before permanent archiving in Zenodo; errors are corrected openly in a new version, not silently. (1) Two coding errors in how the register text was read are reported in 4.5 with a check made afterwards: gon qualifiers in nine road bearings and six inscription bearings were read as a single compass point, and five stones with two records in the register were counted twice. Corrected, the association is slightly stronger and every sensitivity analysis holds; the locked analysis's numbers stand unchanged, and the abstract and the material section say so. (2) The null results for the compass, the farm and the bridge and road stones are described as no difference detected, not as facts; the weaknesses have a point on the age of the farm sites, and the weakness on the age of the road now says that the association with registered older routes is weak. (3) Earlier research: Ekholm (1950) had already described stones turned towards the road and a re-erection that turned a stone; the objection to Klos is now attributed to Mats G. Larsson (2010), added to the references; the pages for Ekholm and McLeod are given; a reference not cited in the text is removed. (4) The plan's two robustness tests that were not reported (the farm within 500 metres, only Uppland and Södermanland) are in 5.2. (5) Form: the share facing south is described as what it counts (between south-east and south-west), 5.3 says that the sensitivity analyses drew new shuffles, the caption of Figure 1 says what the same compass point means, minus signs, the chart's maker, and under Data and code the corpus versions, commits, fetch scripts and fetch dates for the register.
- **Version 4, 2026-11-05.** A second review before archiving in Zenodo checked what version 3 changed; no numbers of the analysis have changed. (1) The objection to Klos is given more exactly: Larsson (2010) objects on p. 252 that the grave fields are overstated, and on p. 250 he notes that Klos's criteria for an ancient road were more restrictive than in his own earlier survey, not than in earlier research in general. (2) The null result for the compass is described as no consistent sign: 5.2 now explains that half B on its own leans towards the south-west ($p = 0.039$) but neither half A nor the whole material does, and the discussion names why the tests of the compass and the farm say less (the age of the farm sites, and that Rayleigh's test detects only a single preferred direction) instead of calling them weaker without a measure; the opening summary and the essentials box say the same. (3) The share facing between south-east and south-west (32 percent) is compared with the 31 percent an even spread gives. (4) 5.1 says that the 123 with a road are records for 120 stones. (5) In 4.5, half B's p value with each stone counted once is 0.0075, which was printed as 0.007 through a rounding error; it is now 0.008. (6) Data and code gives the commit of the check in 4.5 and the builder of version 3 (6eae0af). (7) Changes that version 3 made but did not log: the weakness on the age of the road no longer says that a later road can only weaken the association, since a road laid past the stones or a stone reset to face a newer road could also strengthen it; 5.2 gives Rayleigh's test in each half; the step with the corrected rule for moved stones says that the main analysis already uses that rule, and 4.4 and 5.3 name

the rule by the date of its correction; the sentence under Data and code about a future release with a DOI is removed.

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Cite as

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